LINUX COMMAND CHEAT SHEET

Purpose: A quick-reference guide for common Linux commands used in system administration, network troubleshooting, and beginner cybersecurity labs. Commands and examples should be run only on systems you own or are authorized to administer.

1. Basic Linux Commands

Linux Command	What It Does	When to Use It
pwd	Displays the current working directory.	Check your current location in the filesystem.
ls	Lists files and directories.	View the contents of a directory.
ls -la	Lists all files, including hidden files, with detailed information.	Inspect permissions, ownership, and hidden files.
cd /path	Changes the current directory.	Navigate to another directory.
mkdir folder	Creates a directory.	Organize files into folders.
touch file.txt	Creates an empty file if it does not exist.	Create a new file quickly.
cp file1 file2	Copies a file.	Make a backup or duplicate.
mv old new	Moves or renames a file.	Organize or rename files.
rm file.txt	Deletes a file.	Remove a file you no longer need; verify the path first.
cat file.txt	Displays file contents.	Read a short text file.

2. File Searching and Text Processing

Linux Command	What It Does	When to Use It
find /home -name "*.txt"	Searches for text files beneath /home.	Locate files by name.
grep "error" file.log	Searches a file for matching text.	Find errors in logs.
grep -i "failed" auth.log	Searches without case sensitivity.	Find matching log entries regardless of capitalization.
head -n 10 file.txt	Displays the first 10 lines.	Preview a file.
tail -n 10 file.log	Displays the last 10 lines.	Inspect recent log entries.
tail -f file.log	Follows new lines as a log changes.	Monitor a log in real time.
less file.txt	Opens a scrollable text viewer.	Read a large file without printing it all at once.
wc -l file.txt	Counts lines in a file.	Count log entries or text lines.

3. Users, Groups, and Permissions

Linux Command	What It Does	When to Use It
whoami	Displays the current username.	Confirm which account is active.
id	Displays user ID, group ID, and group memberships.	Check identity and group membership.
groups	Lists the current user's groups.	Verify group membership.
sudo command	Runs an authorized command with elevated privileges.	Perform administrative tasks when necessary.
chmod 640 file.txt	Sets owner read/write, group read, and no access for others.	Restrict access to a file.

Linux Command	What It Does	When to Use It
chmod u+x script.sh	Adds execute permission for the owner.	Make a script executable by its owner.
chown user:group file.txt	Changes a file's owner and group.	Assign correct file ownership.
passwd	Changes the current user's password.	Update your account password.
useradd username	Creates a user account on many Linux distributions.	Create an account with administrator authorization.
groupadd groupname	Creates a group.	Manage shared access for multiple users.

4. Processes and System Monitoring

Linux Command	What It Does	When to Use It
ps	Displays process information.	Inspect running processes.
ps aux	Lists processes across users with details.	Investigate processes and resource usage.
top	Shows a live view of processes and system usage.	Monitor CPU and memory use.
free -h	Displays memory usage in readable units.	Check available RAM.
df -h	Displays filesystem disk space.	Check whether a filesystem is running out of space.
du -sh folder	Summarizes a directory's disk usage.	Identify large directories.
uname -a	Displays kernel and system information.	Identify the running Linux kernel.
uptime	Displays uptime and load averages.	Check how long the system has been running.
kill PID	Sends a signal to a process.	Stop a process you own or administer.

5. Networking Commands

Linux Command	What It Does	When to Use It
ip a	Displays network interfaces and IP addresses.	Find a VM's IP address.
ip route	Displays the routing table.	Identify the default gateway and routes.
ping 192.168.56.10	Tests reachability using ICMP echo requests.	Check connectivity between lab VMs.
traceroute example.com	Shows the network path toward a destination.	Investigate routing issues; some hops may be filtered.
ss -tulpn	Lists listening TCP/UDP sockets and, where permitted, process details.	Identify services listening on the system.
nslookup example.com	Queries DNS for a domain.	Troubleshoot name resolution.
dig example.com	Performs detailed DNS queries.	Inspect DNS records and responses.
ip neigh	Displays the local IP neighbor cache.	Inspect IP-to-link-layer address mappings.
curl -I https://example.com	Requests HTTP response headers.	Check a web server's response headers.
ssh user@192.168.56.10	Opens an SSH connection to a remote host.	Access a lab machine securely.

6. Logs, Services, and Scheduled Tasks

Linux Command	What It Does	When to Use It
ls /var/log	Lists common system log files.	Find logs during troubleshooting.
journalctl	Displays systemd journal logs.	Investigate system and service events.
journalctl -b	Displays logs from the current boot.	Investigate startup issues.
systemctl status ssh	Shows the status of the SSH service on systems where it	Check whether the service is running.

Linux Command	What It Does	When to Use It
	is named ssh.	
systemctl list-units --type=service	Lists loaded service units.	Review system services.
crontab -e	Edits the current user's cron schedule.	Schedule recurring tasks.
crontab -l	Lists the current user's cron jobs.	Verify scheduled jobs.
date	Displays the system date and time.	Check timestamps and scheduling.

7. Archives, Help, and Package Management

Linux Command	What It Does	When to Use It
tar -czf backup.tar.gz folder/	Creates a compressed archive.	Back up project files or logs.
tar -xzf backup.tar.gz	Extracts a gzip-compressed archive.	Restore archived files.
man ls	Opens the manual page for ls.	Learn command options and syntax.
command --help	Displays brief help for many commands.	Quickly check command syntax.
which python3	Shows the executable path found for a command.	Check whether a command is on PATH.
sudo apt update	Refreshes package lists on Debian/Ubuntu-based systems.	Refresh package information before installing software.
sudo apt install nmap	Installs Nmap on Debian/Ubuntu-based systems.	Install a networking tool for authorized lab use.

8. Linux Permission Quick Reference

Permissions are assigned to the owner, group, and others. For regular files, read means view content, write means modify content, and execute means run the file. For directories, execute allows traversal.

Number	Permission	Meaning
7	rwx	Read, write, execute
6	rw-	Read and write
5	r-x	Read and execute
4	r--	Read only
0	---	No permissions

Example: chmod 750 shared-folder gives the owner full access, the group read/traverse access, and others no access.

Use chown user:group file to change ownership. Avoid broad permissions such as chmod -R 777. Do not change SUID bits on system binaries.

9. Cybersecurity Lab Quick Reference

Command	Purpose
ip a	Record each VM's IP addresses.
ip route	Record the default gateway and routes.
ping <lab-IP>	Test reachability between lab VMs.
ss -tulpn	Inspect listening services and ports.
ps aux	Review running processes.
find /home -type f	Practice locating files.
grep -R "pattern" /path	Search files recursively for a pattern.
crontab -l	Check the current user's scheduled jobs.
nslookup example.com / dig example.com	Check DNS resolution.